Electric Drive System Characteristics

- The drive system is an electro-mechanical system, which has electro-mechanical component; Electric Motor and mechanical component; Industrial machinery as mechanical load of electric drive system.
- Drive System Characteristics are divided into static and dynamic characteristics. Static characteristics express the operation of the drive system in steady-state stable mode whereas dynamic characteristics express the operation of the drive system in transient conditions.

A) Electric Motors Static Characteristics are shown in figure (2-1). There are two types: 1- Torque-Speed (Mechanical) characteristic is a relationship of angular speed (ω) or rotational speed (n) as a function of motor electro-magnetic torque (T). 2- Electro-Mechanical characteristic is a relationship of angular speed (ω) or rotational speed (n) as a function of motor current (I).

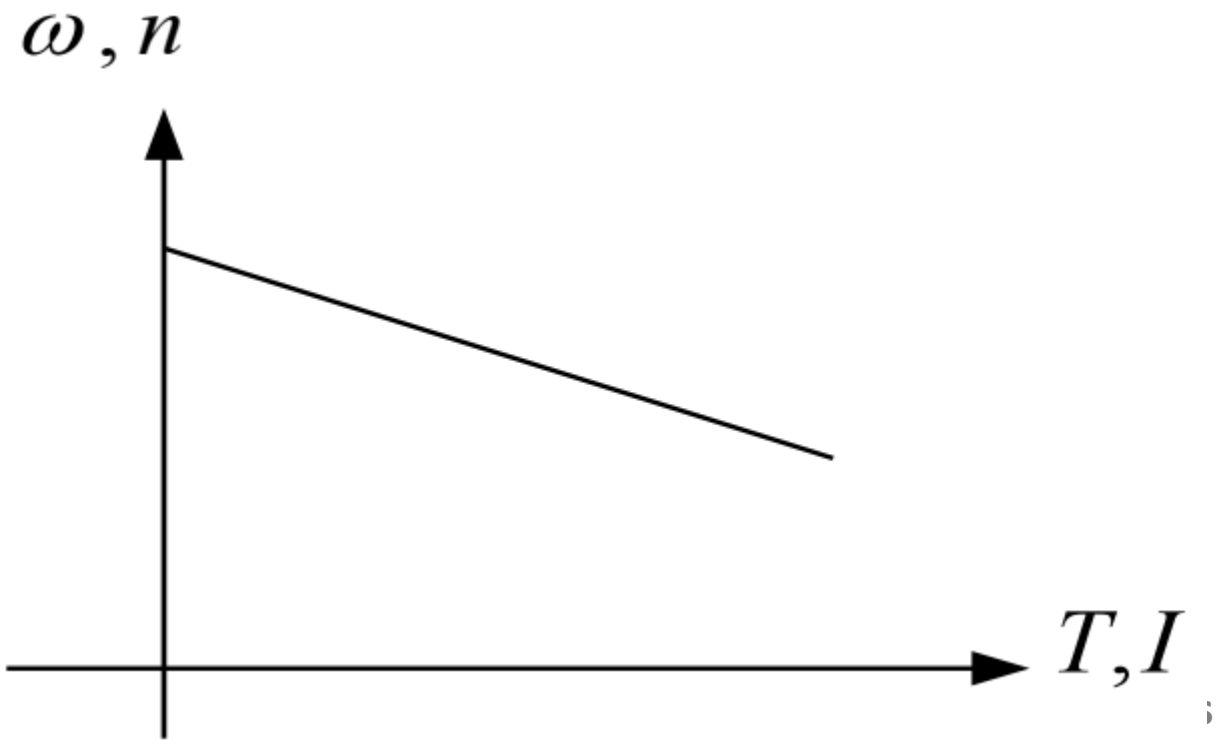


Figure (2-1) Electric motor static characteristics

The electric motor torque speed and Electro-Mechanical static characteristics could be characterized as: 1- Natural characteristics, where the electric motor operates under nominal (rating) parameters without adding resistance to motor circuits. 2- Regulated characteristics, where an electric motor operates under different conditions when varying one motor parameter or adding resistance to motor circuits. Figure (2-2) shows regulated characteristics related to the natural characteristic of separately excited dc motor by varying armature voltage.

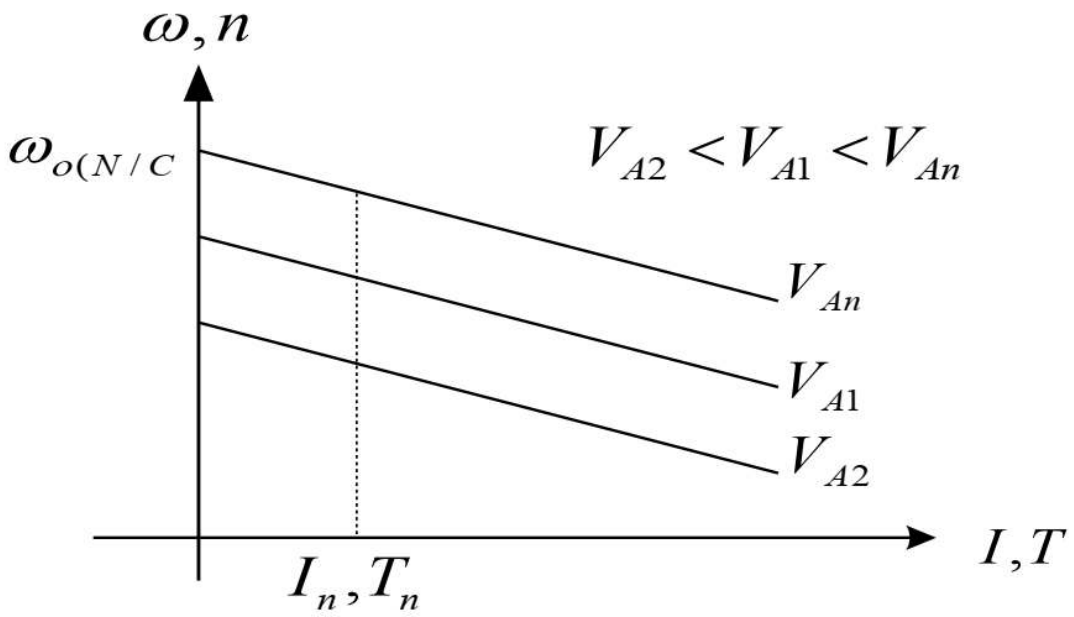


Figure (2-2) Separately excited dc motor regulated characteristics

B) Electric motors dynamic characteristics. It expresses the operation of the drive system in dynamic states (transient states), such as the starting and braking states, reversing or changing any of the system parameters, for example voltage change or load change. It is a time- response of motor parameters relationship, as shown in Figure (2-3).

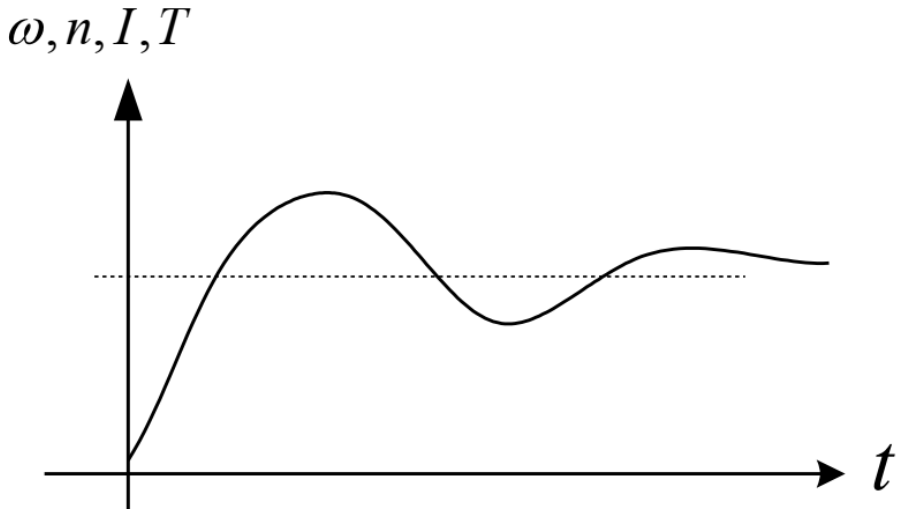


Figure (2-3) Time-response of motor parameters

- The transient state is studied using the motion equation:

$$T = \mathrm{T}_{st} + \mathrm{T}_{dyn} = \mathrm{T}_{l} + J\frac{\mathrm{d}\omega_m}{dt}$$

When :-

1- $(T_L = T)$ **i.e. the motor torque is equal to the load torque, this means from the motion equation relationship that the angular acceleration** $(\frac{\mathrm{d}\omega}{\mathrm{dt}} = 0)$**, that is, the system is operating in a static-state mode at a constant speed** $(\omega = \mathrm{Const.})$

2- If it is $(\mathrm{T_L} < \mathrm{T})$ **, then** $\left(\frac{\mathrm{d}\omega}{\mathrm{dt}} > 0\right)$ **, the speed changes with the change of time and the system operates in a dynamic state with acceleration.**

3- If it is $(\mathrm{T_L} > \mathrm{T})$ **, then** $\left(\frac{\mathrm{d}\omega}{\mathrm{dt}} < 0\right)$ **, speed changes with the change of time and the system operates in a dynamic state with deceleration.**

Hardness of Electric Motors Torque-Speed Characteristic

- Torque-speed characteristic hardness for graphical representation is the relationship of a change in torque with a change in speed.
- The hardness relationship is: $$\beta = \frac{\Delta T}{\Delta \omega}$$
- The torque-speed characteristics of electric motors are divided into three types according to their hardness: 1- Absolutely hard torque-speed characteristic: It remains at a constant speed regardless of the torque increase, an example of which is the synchronous motor characteristic, as shown in Figure (2-8).

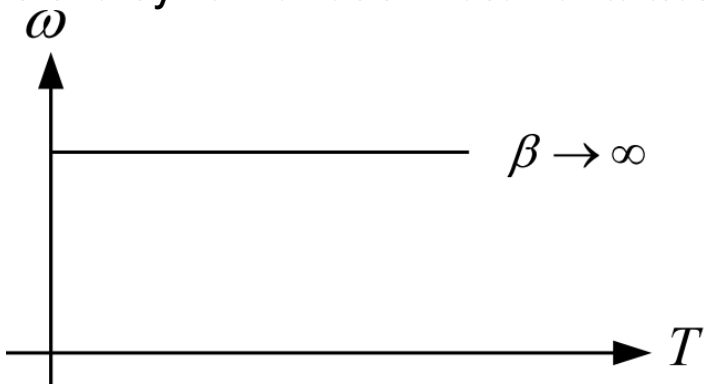


Figure (2-8) The characteristic of absolute hardness

- 2- Hard torque-speed characteristic: Hard torque- speed characteristics are the characteristics of separately excited dc motor and three-phase asynchronous motors in which the speed decreases slightly with increasing torque, as shown in Figure (2-9).

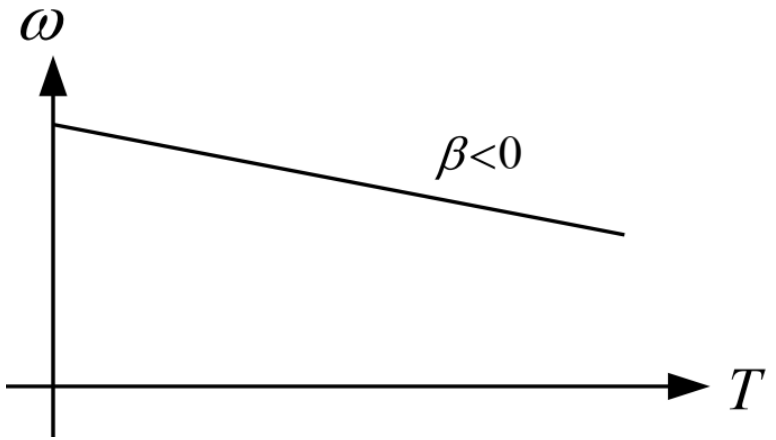


Figure (2-9) Hard torque-speed characteristic

- 3- Soft torque-speed Characteristic: soft torque-speed characteristics are the characteristics of series- field dc motors and compound field dc motors in which the speed decreases dramatically (بشكل كبير) with increasing torque, as shown in Figure (2-10).

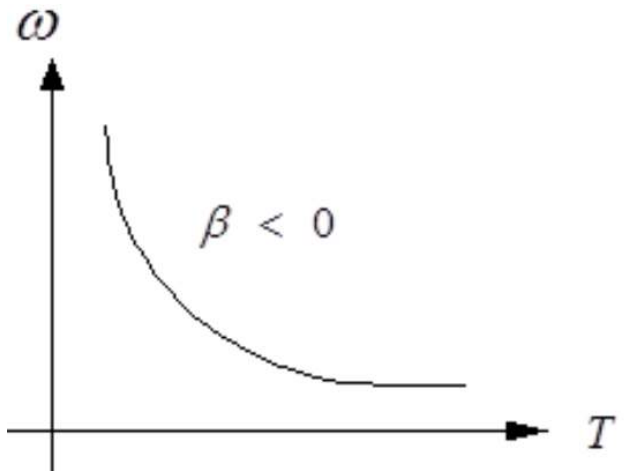


Figure (2-10) Soft torque-speed characteristic

Drive System Torque-Speed Characteristics

- They are the combined Torque-Speed Characteristics of the motor and the machinery. The relationship between speed and torque is represented as shown in Figure (2-11),

whereas: 1- Torque-speed characteristic of the machinery 2- Torque-speed characteristic of separately excited dc motor. The drive system operates in steady-state mode at the intersection of the torque-speed characteristics of the motor and the machinery at operating point (A), which means

$$\omega = \omega_L \text{ and } T = T_L .$$

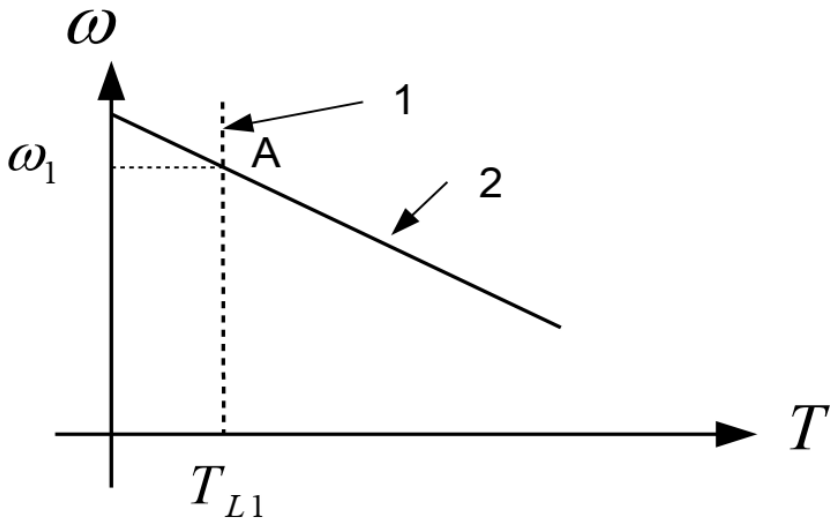


Figure (2-11) Drive system torque-speed characteristics

- The stability of the drive system (stable operation) depends on the relationship between the hardness of the torque-speed characteristic of the motor to the hardness of the torque-speed characteristic of the load. Figure (2-12) shows torque-speed characteristics of different hardness.

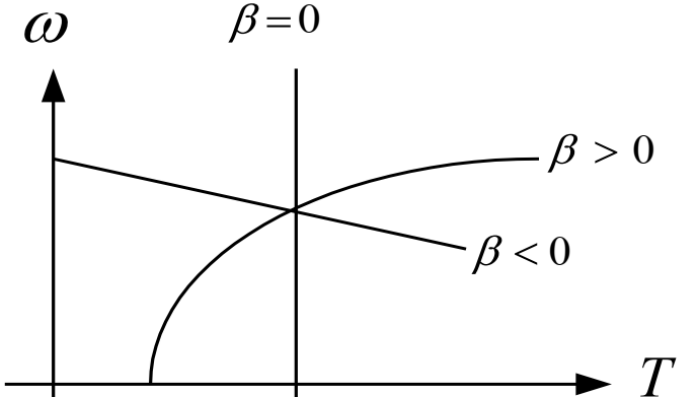


Figure (2-12) Torque-speed characteristics of different hardness

The importance of torque-speed characteristics

1- Determine the loading mode of electric motor that depends on the location of the intersection between the torque-speed characteristics of the motor and the machinery. The loading modes of electric motors will be recognized when studying the static (steady-state) characteristics of electric motors.

2- Study the dynamic (transient-mode) characteristics of the motor. The initial and final values (Final or steady-state value) of the variable are determined when operating on the same characteristic or when moving (cross over) from one characteristic to another.

3- Determine the hardness of torque-speed characteristics.

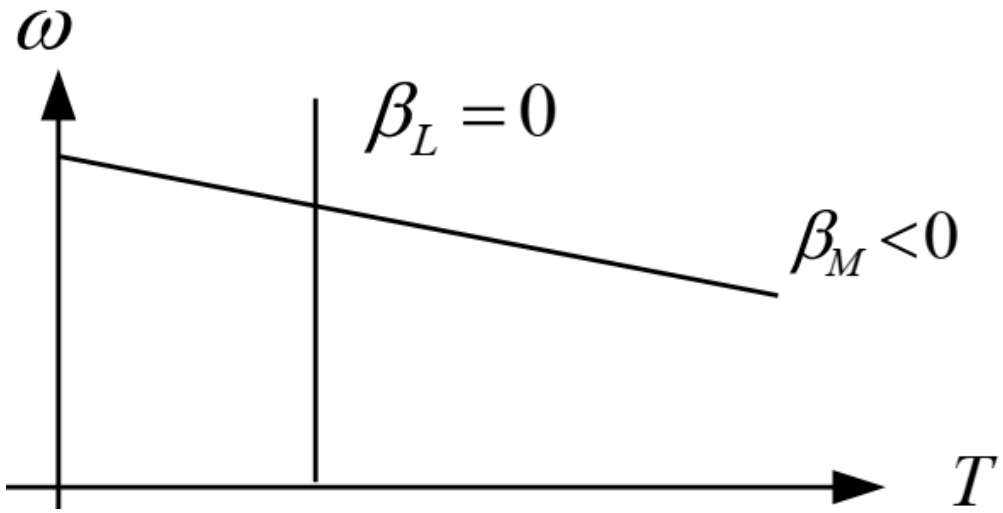
4- Judging (determine) the stability of the system by comparing the hardness of the motor torque-speed characteristic (β_M) and the hardness of torque-speed characteristic of the load (β_L), and in order for the system to be stable, the hardness of the motor torque-speed characteristic must be less than the hardness of the load (machinery) and this is the condition for the stability of the drive system $\beta_M < \beta_L$.

STEADY STATE STABILITY

- In the case of a graphical representation of torque-speed characteristics:

$$\beta_M = \frac{\Delta T_M}{\Delta \omega_M},\ \beta_L = \frac{\Delta T_L}{\Delta \omega_L}$$

The figure below represents a stable operating drive system that meets the stability conditions



- In the case of mathematical representation of motor and load torque-speed characteristics as follows:

$$T = A + B\omega \qquad (1) \qquad T_L = C + D\omega_L \qquad (2)$$

The hardness of the motor and load torque-speed characteristics are calculated by deriving (1) and (2) as follows:

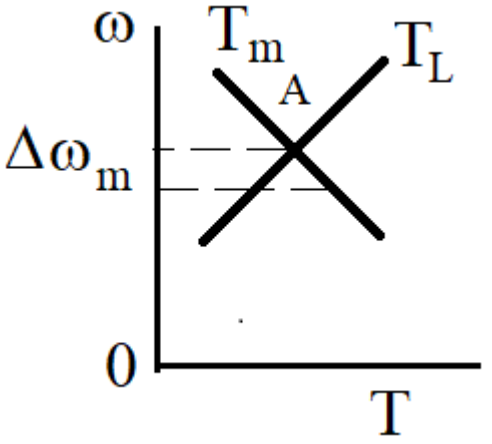
$$\beta_m = \frac{dT}{d\omega} \qquad (3) \qquad \beta_L = \frac{dT_L}{d\omega_L} \qquad (4)$$

The condition for stability of the drive system is $\beta m < \beta_L$, or

$$\frac{dT}{d\omega} < \frac{dT_L}{d\omega_L}$$

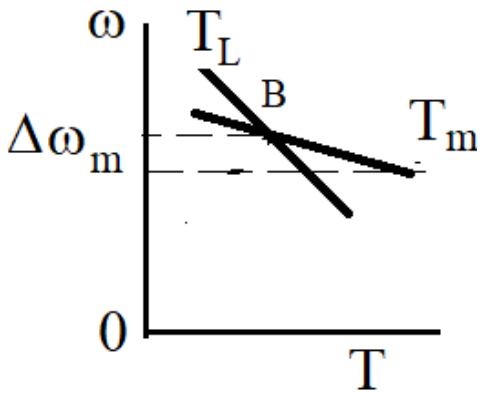
→ Equilibrium speed of a motor-load system is obtained when motor torque equals the load torque.
→ The drive system will operate in a steady - state at this speed which is known as the speed of stable equilibrium.
→ The concept of steady-state stability has been developed to readily evaluate the stability of an equilibrium point from the steady-state speed-torque curves of the motor and load, thus avoiding the solution of differential equations valid for the transient operation of the drive.
→In most drives, the electrical time constant of the motor is negligible compared to its mechanical time constant.
→Therefore, during transient operation, the motor can be assumed to be in electrical equilibrium implying that steady-state speed-torque curves are also applicable to the transient operation.
→Let us examine the steady-state stability of equilibrium points with some of the following examples.

Example:
→Consider a motor-load system of motor torque, and load torque, as shown in the aside figure.



→The equilibrium point will be termed as stable when the operation will be restored to it after a small departure from it due to a disturbance in the motor or Load.
→Let the disturbance cause a reduction of $\Delta\omega_m$ in speed.
→At new speed, if the motor torque is greater than the load torque, consequent, motor will accelerate and operation will be restored to A.
→Similarly, an increase of $\Delta\omega_m$ in speed caused by a disturbance will make load torque greater than the motor torque, resulting in deceleration and restoration of operation to Point "A".
→Hence the drive is steady-state stable at Point A.

→Let us consider equilibrium point B' which is obtained when same motor drives another load as shown aside.



→The equilibrium point will be termed as stable when the operation will be restored to it after a small departure from it due to a disturbance in the motor or Load.

→Let the disturbance causes a reduction of $\Delta\omega_m$ in speed.

→At new speed, motor torque is greater than the load torque, consequent, motor will decelerate and operation will be restored to A.

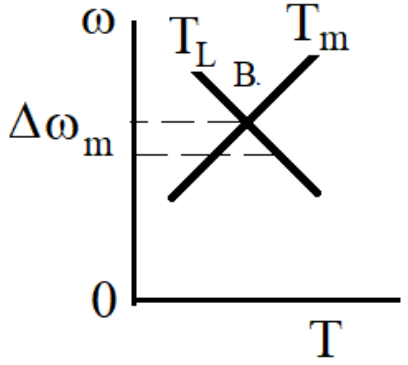
→Similarly, an increase of $\Delta\omega_m$ in speed caused by a disturbance will make load torque greater than the motor torque, resulting into deceleration and restoration of operation to Point "A".

→Hence the drive is steady-state stable at Point A.

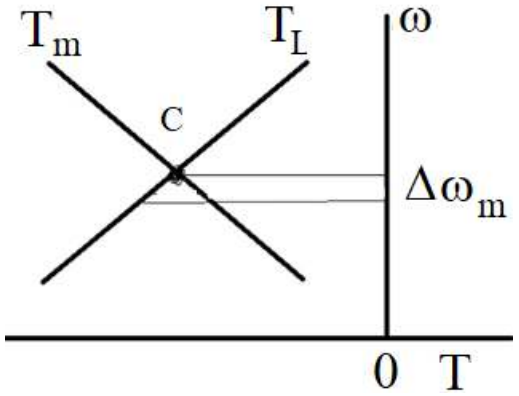
→a $\Delta\omega_m$ decrease in speed caused the load torque to become greater than the motor torque, the drive decelerates and the operating point moves away from "B".

→similarly when working at "B" an increase in steed will make motor torque greater than the load torque, resulting in acceleration of the drive which will move the operating point away from B.

→Thus B" is an unstable point of equilibrium.



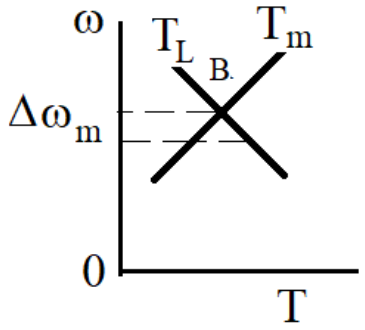
→Let us consider another equilibrium Point "C" whose steady-state characteristics as shown below.



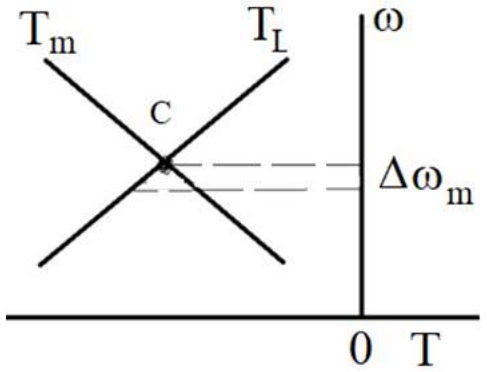
→a ΔΩm decrease in speed caused the load torque to become greater than the motor torque, the drive decelerates and the operating point moves away from "B".

→similarly when working at "B" an increase in steed will make motor torque greater than the load torque, resulting in acceleration of the drive which will move the operating point away from B.

→Thus B" is an unstable point of equilibrium.

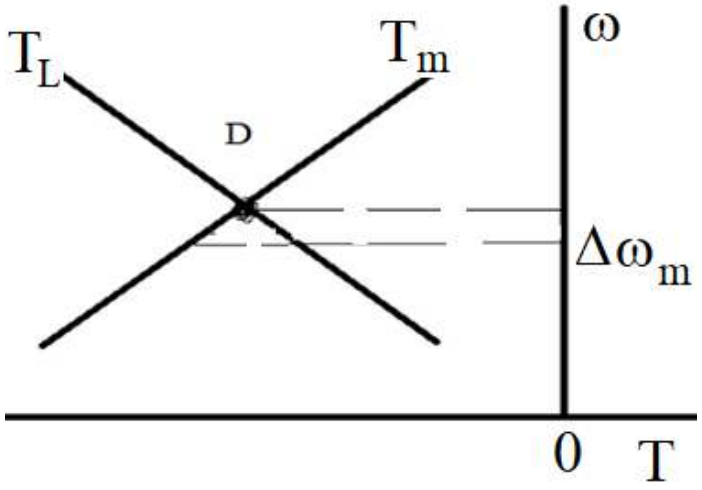


→Let us consider another equilibrium Point "C" whose steady-state characteristics as shown below.



→A decrease in speed causes motor torque to be negatively greater than load torque, results accelerates the drive and the operating point moves to "C".
→Similarly, an increase in speed causes the motor's torque to be negatively less than the load torque, resulting in deceleration of the drive and the operating point moving to "C"
→Thus "c" is a stable point of equilibrium.

→Let us consider another equilibrium Point "D" as shown below.



→ A decrease in speed, results load torque negatively greater than motor torque results decelerate the drive and operating point moved away from D.
→ Similarly, an increase in speed, results load torque is negatively, Smaller than motor torque results accelerated the drive and operating point moves away from D.
→Thus, D is an unstable point of equilibrium.
→from above discussion, "An equilibrium Point will be stable when an increase in speed cause load - torque to exceed the motor torque " (or) "An equilibrium point will be stable when an decrease in speed causes motor-torque to exceed the load torque.
i.e, at equilibrium point the drive satisfy the condition.

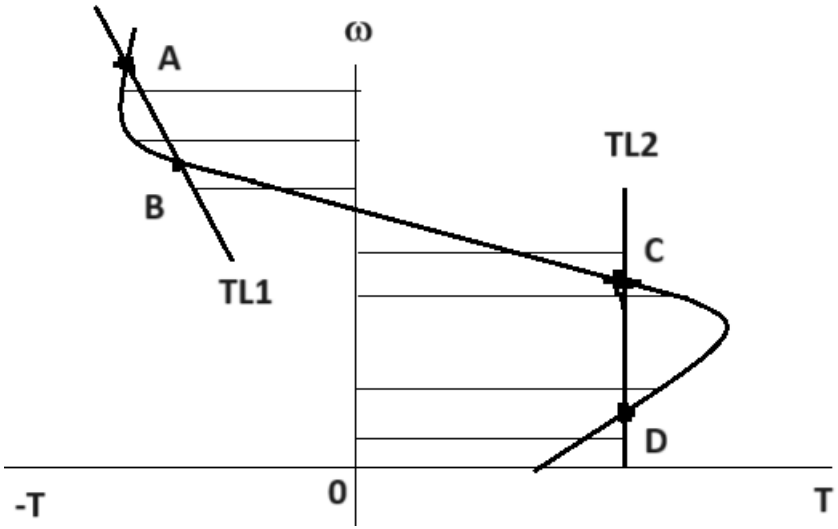
$$\frac{dT_l}{\mathrm{d}\omega_m} > \frac{dT}{\mathrm{d}\omega_m}$$

→"In a drive "An equilibrium point will be stable when an increase in speed causes load torque to exceed the motor torque" [or] "An equilibrium point will be stable when a decrease in speed motor causes motor torque to exceed the load torque i.e at equilibrium

$$\frac{dT_l}{\mathrm{d}\omega_m} > \frac{dT}{\mathrm{d}\omega_m}$$

Examples

- EX1: Figure below shows plots of speed vs motor and load torques. Comment on the stability of the operating points A, B, C and D.



Solution:

- **Stability of operating Point A**
- → If there is any speed increase at Point "A" motor Torque (T) negatively greater than load torque (TL) hence the drive accelerates and operating point moving away from "A.
- →if there is any speed decrement at point "A" the motor Torque (T) negatively smaller than load torque (TL) hence the drive decelerates and operating point moving away from "A".
- →hence operating point A is unstable.

Stability of operating Point "B"

- → If there is any speed increment at Point "B" the load torque (TL) is greater than motor torque (T) hence the drive is decelerate and operating point moving towards "B".
- → If there is any Speed decrement at Point "B" the Load torque (TL) is less than motor torque (T) hence the drive is accelerate and operating point moving towards "B".
- →hence the operating point "B" is stable.

Stability of operating point "C".

- →If there is any speed increment at Point "C" the load torque (TL) is greater than motor torque (T), hence the drive is decelerating and operating point moving towards "C".
- →If there is any speed decrement at Point "c" the load torque (TL) is less than motor torque (T), hence the drive is accelerating and operating point moving towards "C".
- →hence the operating point "C' is Stable.

Stability of operating Point "D".

- → if there is any speed increment at point "D", the motor torque (T) is seater than load torque and (TL) and the drive accelerates and the operating point moves away from "D".
- → if there is and speed decrement at point "D" the motor torque (T) is less than load torque (TL) and the drive decelerates and the operating point moves away from "D".
- →hence operating point "D" is unstable.
- →for the given drive operating points "A" & "D" are unstable and operating points "B"&" C" are Stable.

EX2:

A drive has following equations for motor and load torques

$$T = (1 + 2\omega m) \text{ and } T_l = 3\sqrt{\omega_m}$$

obtain the equilibrium points and determine their steady-state stability.

Solution:

- At equilibrium Points motor torque = load torque $T = T_l$

$\rightarrow 1 + 2\omega_m = 3\sqrt{\omega_m}$

- Squaring on both sides
- $(1 + 2\omega_m)^2 = (3\sqrt{\omega_m})^2$

$$4\,\omega_m{}^2 + 4\omega_m + 1 = 9\omega_m$$
$$4\,\omega_m{}^2 - 5\,\omega_m + 1 = 0$$
$$4\,\omega_m{}^2 - 4\omega_m - \omega_m + 1 = 0$$
$$4\omega_m(\omega_m - 1) - (\omega_m - 1) = 0$$
$$(4\,\omega_m - 1)(\omega_m - 1) = 0$$

$4\omega_m - 1 = 0 \rightarrow \omega_m = \frac{1}{4}$ rad/sec

$\omega_m - 1 = 0 \rightarrow \omega_m = 1$ rad/sec

- At equilibrium Points motor torque = load torque $T = Tl$
- equilibrium Consists at "2" Points with $\omega m = \frac{1}{4}$ and $\omega m = 1$ rad/sec.
- If $\omega m = \frac{1}{4}$; $T = 1 + 2\left(\frac{1}{4}\right) = 1.5$ N-m
- if $\omega m = 1$; $T = 1 + 2(1) = 3$ N-m
- equilibrium points are A={1.5, 0.25) and B = (3,1).
- **Stability at point A:**

$$\beta_m = \frac{d\,T}{d\omega_m} = 2\,, \qquad \beta_l = \frac{dT_l}{\mathrm{d}\omega_m} = \frac{3}{2 \times \sqrt{\omega_m}} = \frac{3}{2 \times \sqrt{0.25}} = 3$$

Since βl >βm , So Point is stable

Stability at point B:

$$\beta_m = \frac{d\,T}{d\omega_m} = 2, \qquad \beta_l = \frac{dT_l}{\mathrm{d}\omega_m} = \frac{3}{2\times\sqrt{\omega_m}} = \frac{3}{2\times\sqrt{1}} = 1.5$$

- Since βm > βl , So Point is unstable